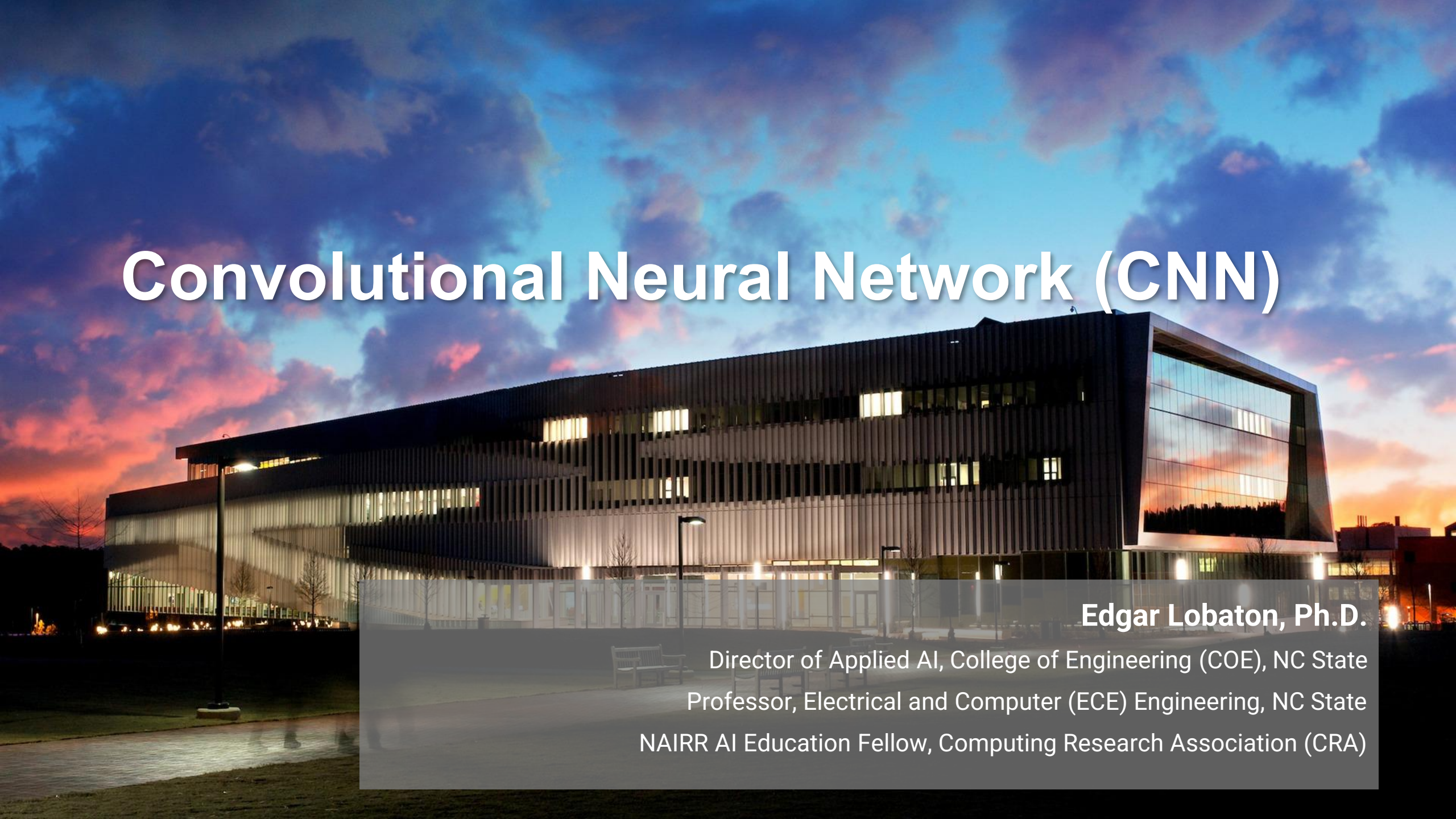


Convolutional Neural Network (CNN)

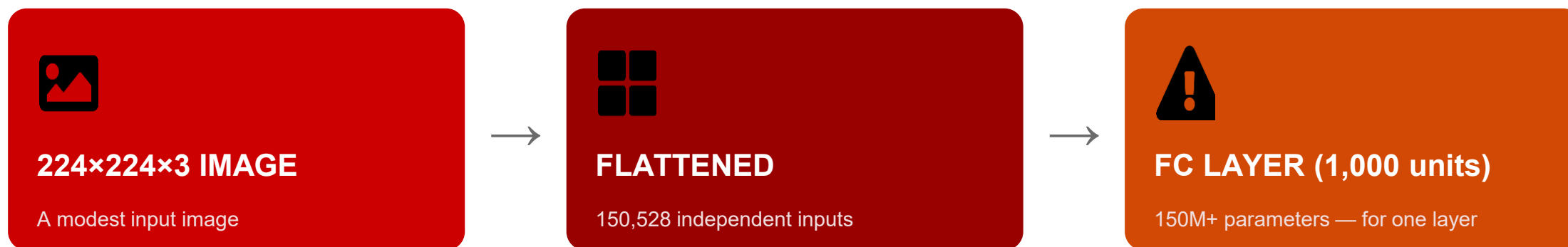
A photograph of a modern, multi-story building with a large glass facade, illuminated from within and reflecting the vibrant colors of a sunset sky. The sky is filled with clouds in shades of blue, orange, and pink. The building's architecture features a mix of dark and light panels, and the interior lights are visible through the windows.

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Why Fully-Connected Layers Fail on Images

The problem isn't accuracy — it's scale and structure

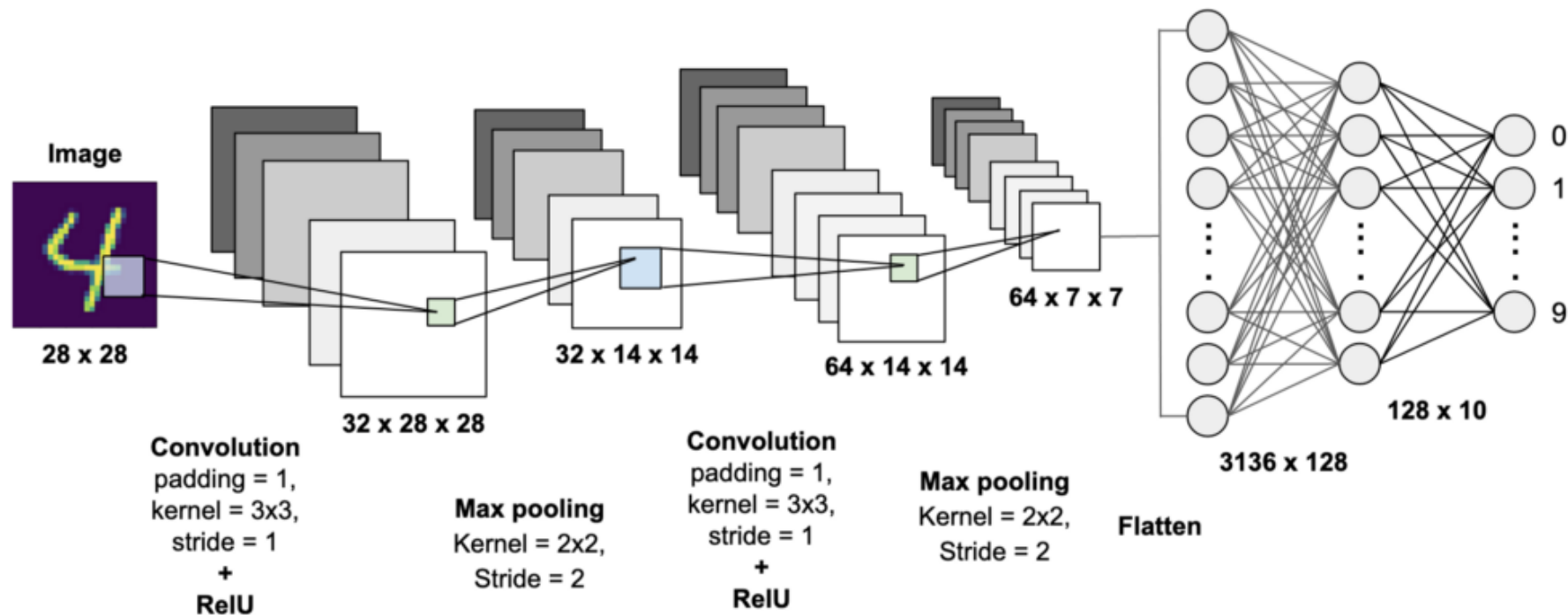


And parameter count is only half the problem: flattening also throws away spatial structure — nearby pixels are treated as completely independent, when they're highly related.

Convolutional layers fix both problems at once: far fewer parameters, and structure that respects spatial relationships.

The Architecture

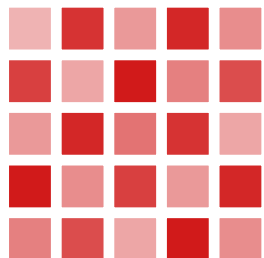
- Main Components: **Convolutional Layer** (convolution plus activation function), **Pooling** (subsampling), and Fully Connected Layers (i.e., an MLP)



The Core Operation: Convolution

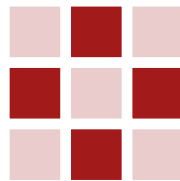
A small learned filter slides across the image, computing a local response

INPUT PATCH



*

KERNEL (3×3)



=

FEATURE MAP



Here, a vertical-edge kernel produces a strong response exactly where a vertical edge appears in the patch.



Local Receptive Fields

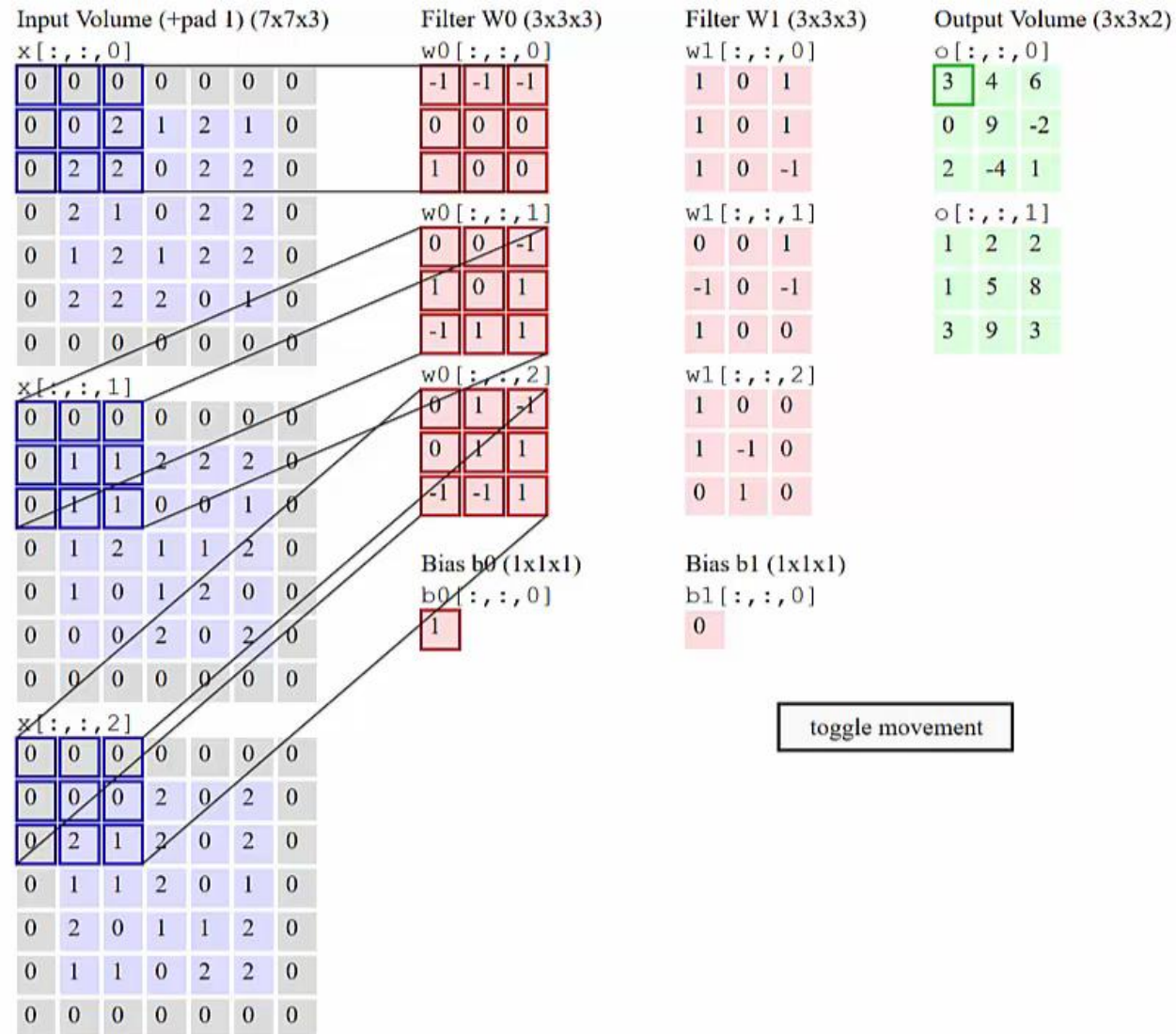
Each output value depends only on a small neighborhood — not the entire image.



Weight Sharing

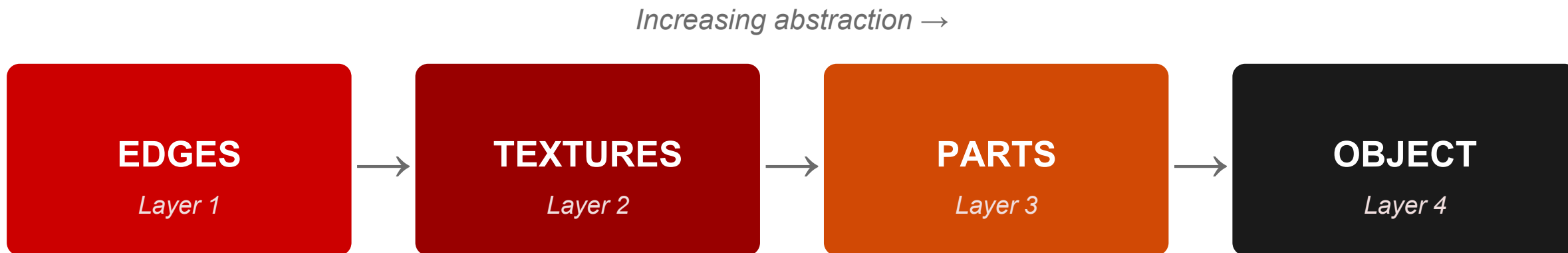
The same filter is reused at every position, instead of learning a separate weight per pixel.

The Core Operation: Convolution



Depth & Hierarchy

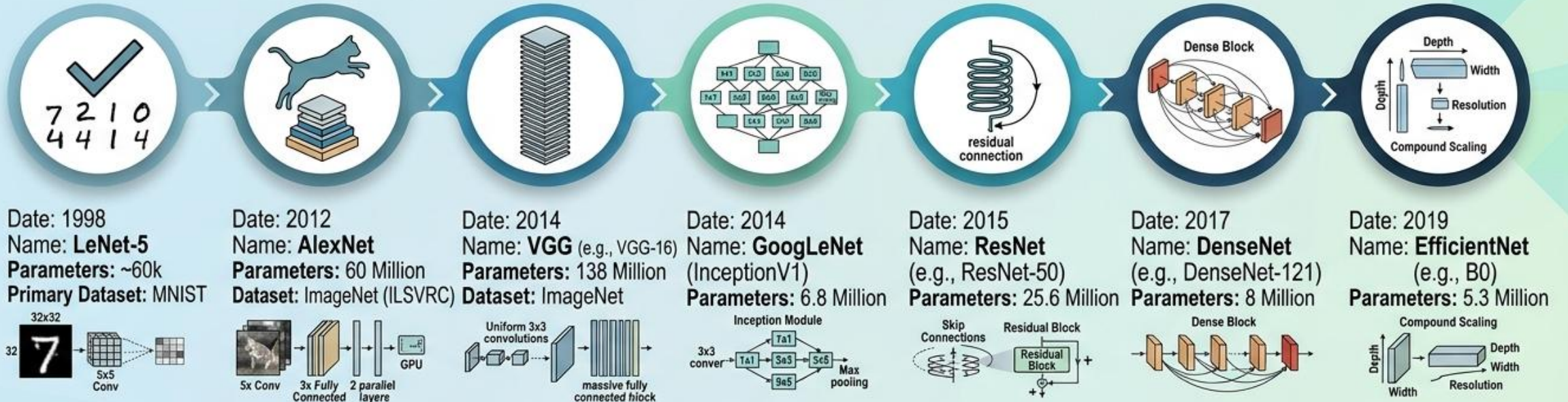
Stacking convolutional layers builds increasingly abstract representations



This is the same abstraction idea from the Deep Learning overview slide — now made concrete with an actual operation behind it.

Architecture Evolution: A Fast Pass

CONVOLUTION-DOMINANT ERA



EMERGING & CONVERGING ERA



CNN Engineering Applications

Where the automatic feature learning actually pays off



Pavement / Bridge Crack Detection

The CNN that finally does the job the hand-built HOG/SVM pipeline struggled with.



Structural Component Recognition

Identifying components in drone imagery, built up from the edges → parts hierarchy.



Thermal / RGB Fusion for Inspection

Combining modalities for rail and infrastructure inspection — your CRISI work.



General Visual Inspection

Any task where the defect is visual but its exact appearance varies too much to hand-code.

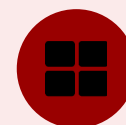
CNN Limitations

The honest counterweight — and the opening for what comes next



Needs Large Labeled Datasets

Strong performance typically requires substantial labeled training data.

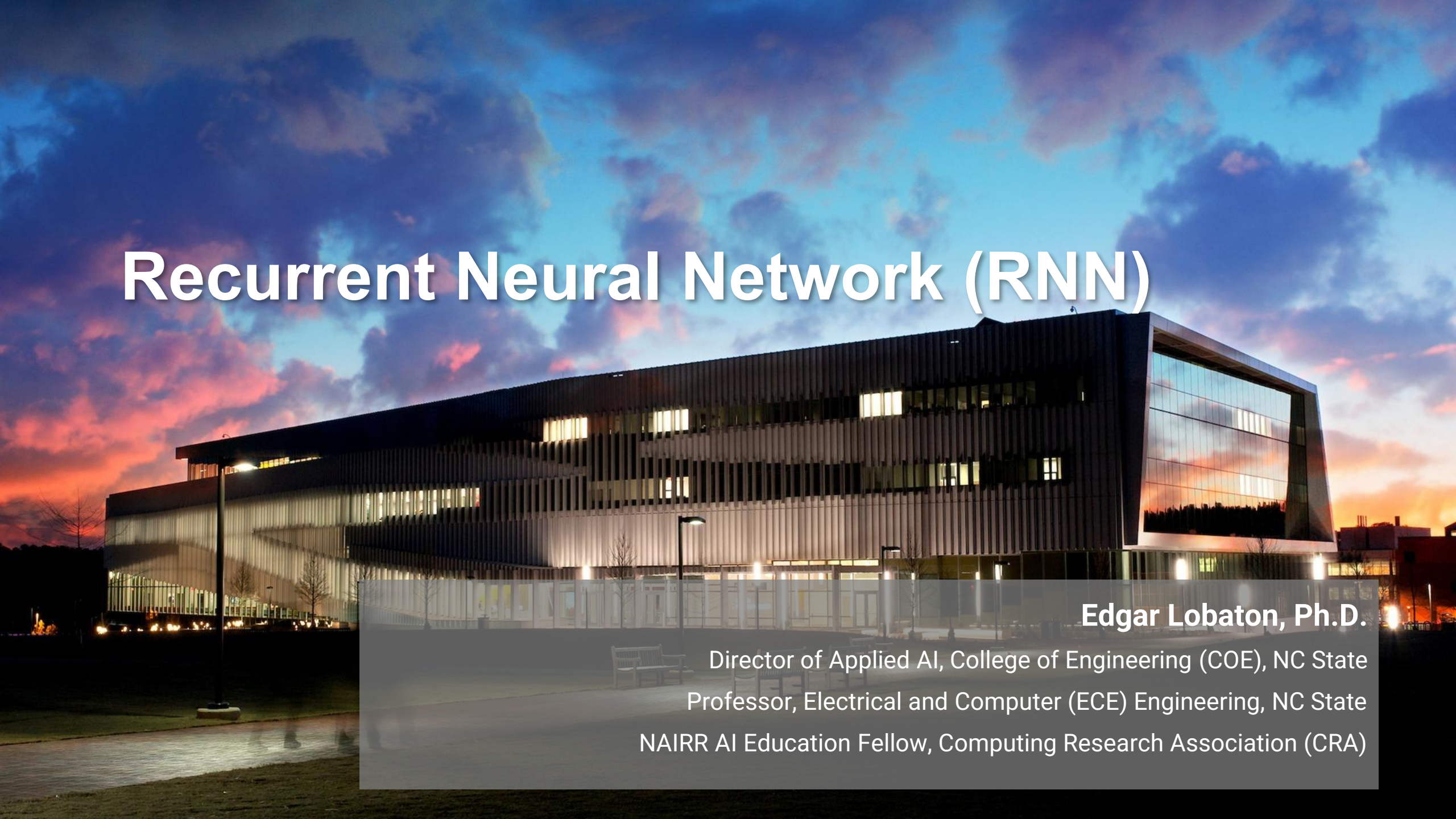


Assumes Fairly Fixed Input Structure

Built around a spatial grid — not naturally suited to variable-length data.

No native way to handle sequences: a single image, yes. A time-series of sensor readings, or a video, not natively — that's where recurrent networks come in.

Recurrent Neural Network (RNN)

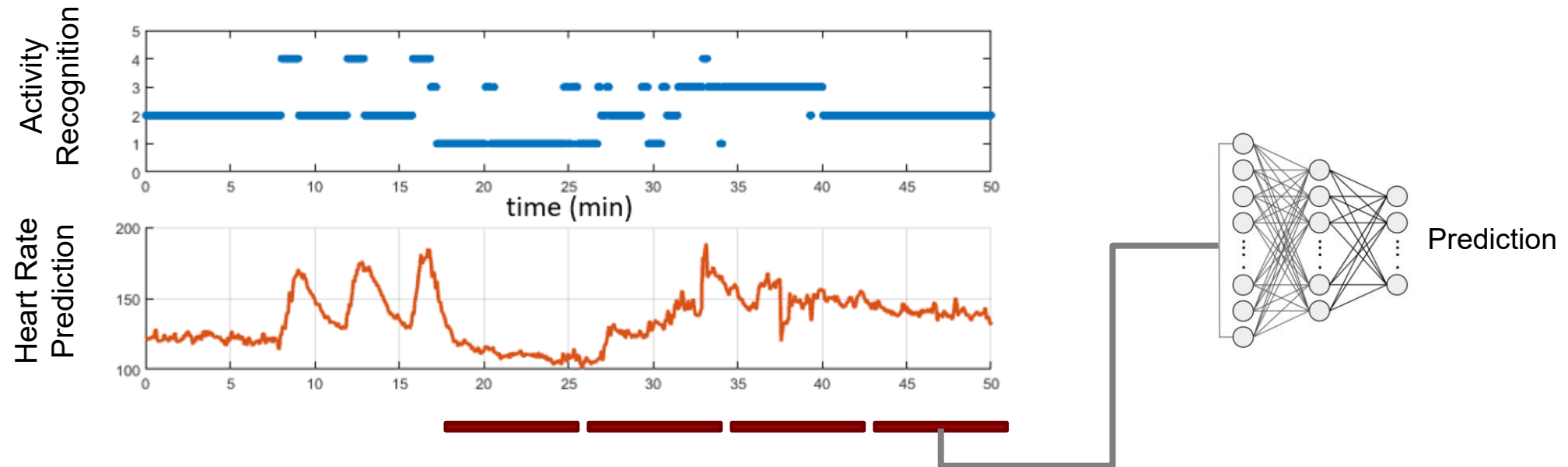
A photograph of a modern, multi-story building with a large glass facade, reflecting the vibrant colors of a sunset sky. The building is illuminated from within, and the sky is a mix of blue, orange, and pink. The title 'Recurrent Neural Network (RNN)' is overlaid in white text on the upper left.

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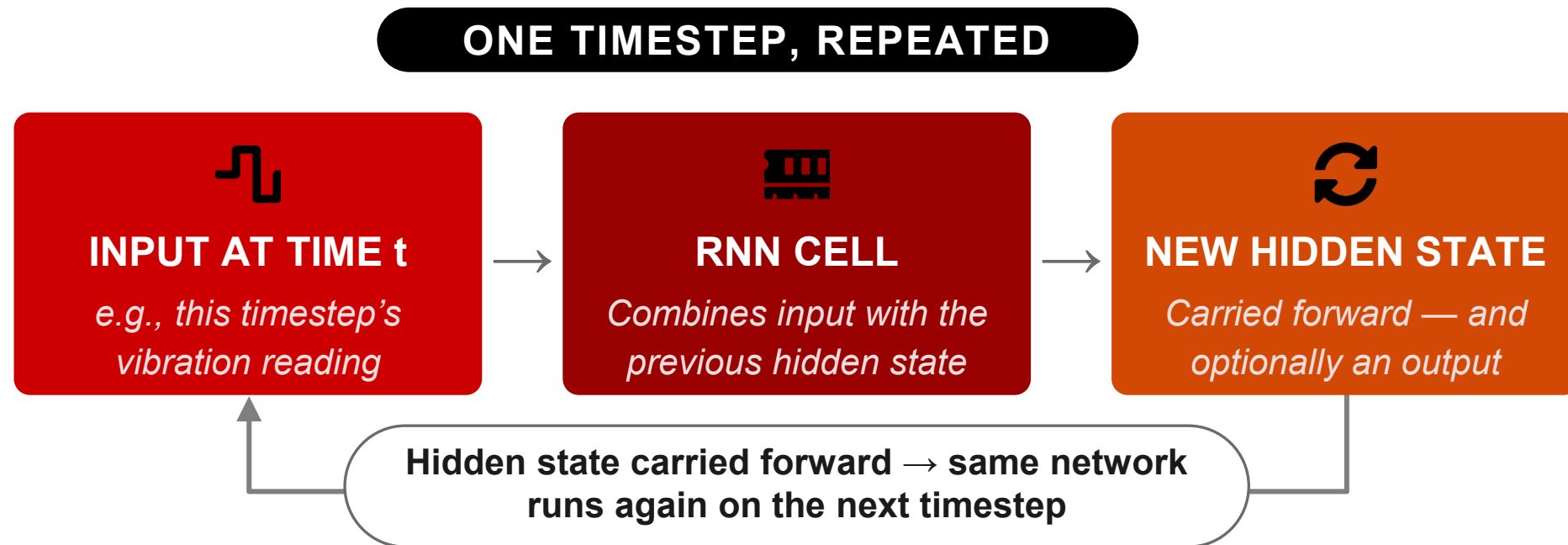
Predictive Models for Sequential Data

- We need something more specialized for time-series data (e.g., NLP, stock market, physiological responses, video sequences)
- MLPs and CNNs are an alternative. However, they do not exploit potential temporal information. They are memoryless.



The Core Idea: Hidden State Through Time

The same recurrent loop pattern from the LLM session — now carrying a learned state



Parameter sharing across time — the RNN version of the weight-sharing-across-space idea from convolution.

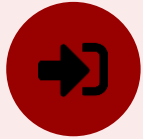
LSTM: Gating as the Fix

Conceptual view — learned gates control what the network remembers



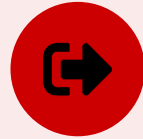
Forget Gate

Decides what to discard from the existing memory.



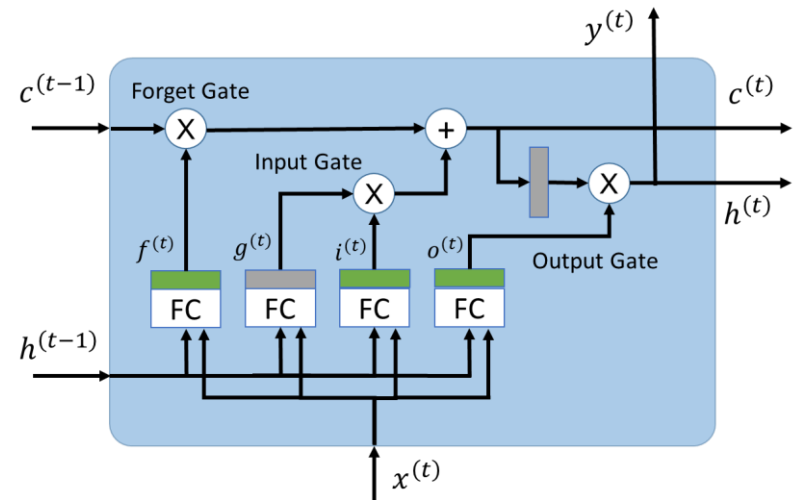
Input Gate

Decides what new information to add to memory.



Output Gate

Decides what part of memory to expose as output.



Cell State — a controllable long-term memory, instead of one that decays automatically

Not the full gate equations here — the takeaway is that gates give the network a choice about what to keep, rather than forgetting by default.

RNN Engineering Applications

Where sequence matters as much as the individual reading



Predictive Maintenance from Time-Series

Vibration or sensor streams where recent history predicts what comes next.



Fatigue / Failure Forecasting

Revisiting the fatigue-life scenario as a sequence problem, not a single-shot fit.



Structural Response Modeling

Modeling how a structure responds over time to a sequence of loads.



Anywhere Order Carries Information

If yesterday's reading changes how you interpret today's, this family applies.

Synthesis: Spatial vs. Sequential Structure

Two families, one underlying idea — build in the right assumption for the data

	CNN	RNN
Exploits	Spatial structure — nearby pixels relate	Sequential structure — nearby timesteps relate
Sharing mechanism	Same filter reused across space	Same cell reused across time
Core building block	Convolution + pooling	Hidden state passed forward
Known weakness	No native handling of sequences	Long-range dependencies decay (vanishing gradient)

Same underlying principle, different axis: share weights wherever the data has a repeating structure to exploit.

AutoEncoder (AE)

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Why Learn Without Labels?

Everything so far has needed labels or a next-token target — what if you just have raw data?

THE SCENARIO

Years of vibration data from healthy pumps — but almost no labeled examples of the faults you actually care about.

THE AUTOENCODER IDEA

Skip labels entirely. Train the network to reconstruct its own input — the input itself is the training target.

This is self-supervised learning: the loss compares the reconstruction to the original input — no human-provided label required.

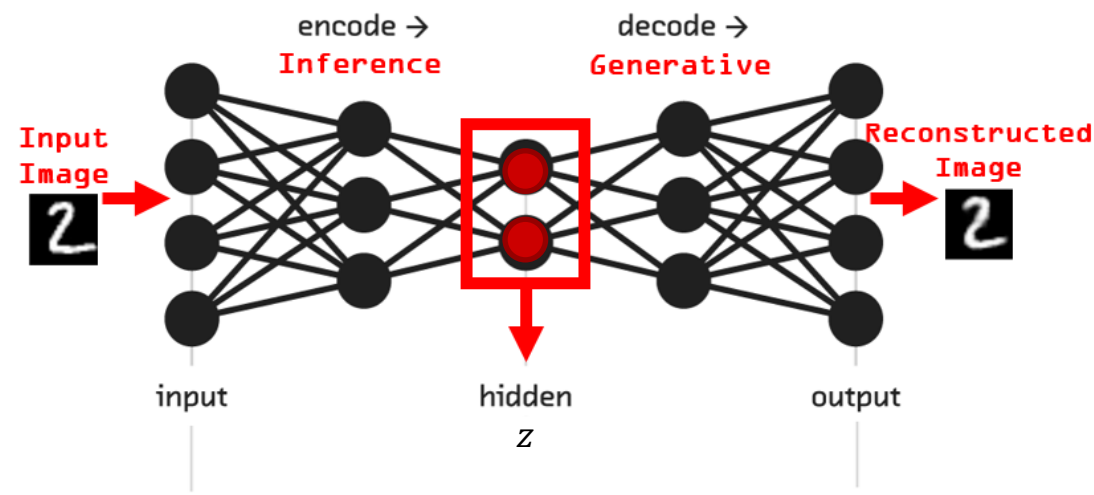
The Core Architecture: Encoder → Bottleneck → Decoder

Trained end-to-end to reconstruct its own input



The reconstruction loss (e.g., mean squared error) compares OUTPUT to INPUT directly — that comparison is the entire training signal.

***Notice the shape: wide → narrow → wide.
That narrow middle is doing all the work.***



Why the Bottleneck Matters

Without a constraint, the network could just learn to copy



NO REAL BOTTLENECK

If the latent code is as large as the input, the network can just learn the identity function — perfect reconstruction, zero useful structure learned.



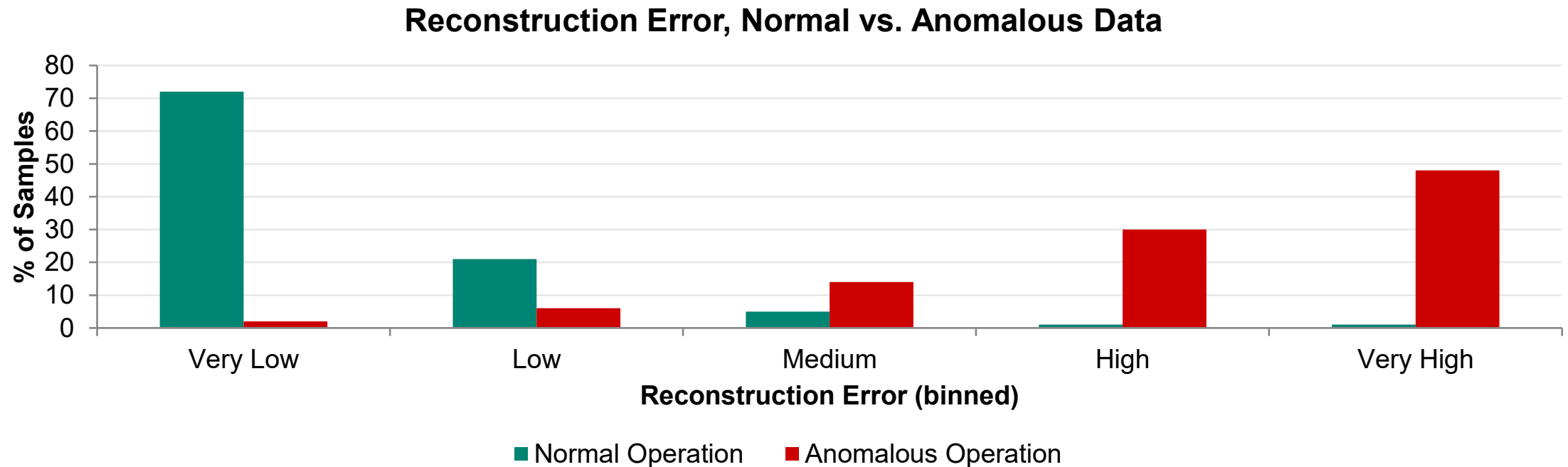
A REAL BOTTLENECK

A latent code far smaller than the input forces the network to discard noise and keep only the structure that matters most for reconstruction.

The bottleneck width is a knob: too wide and it copies, too narrow and it can't reconstruct at all.

Anomaly Detection: The Killer App for Engineers

Train only on normal operation — let reconstruction error do the flagging



Normal samples reconstruct well (low error); the model never learned the anomalous pattern, so it reconstructs poorly (high error) — no fault labels required.

Other Engineering Applications

Beyond anomaly detection



Compression

Encode high-dimensional sensor data into a compact latent code for cheaper storage or transmission.



Denoising

Clean up noisy sensor signals by reconstructing the underlying clean version.



Pretraining / Feature Extraction

Learn a useful representation from unlabeled data, then reuse it as input to a smaller supervised model.

Basic Architectures

<https://forms.gle/XCYrDk1P2nBSHWCU9>



Transformers

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The Gap Transformers Fill

A direct continuation of the RNN limitation from the last session



RNNs (recap)

Process one timestep at a time. Can't parallelize across the sequence, and gradients decay over long distances — even with LSTM/GRU gating.



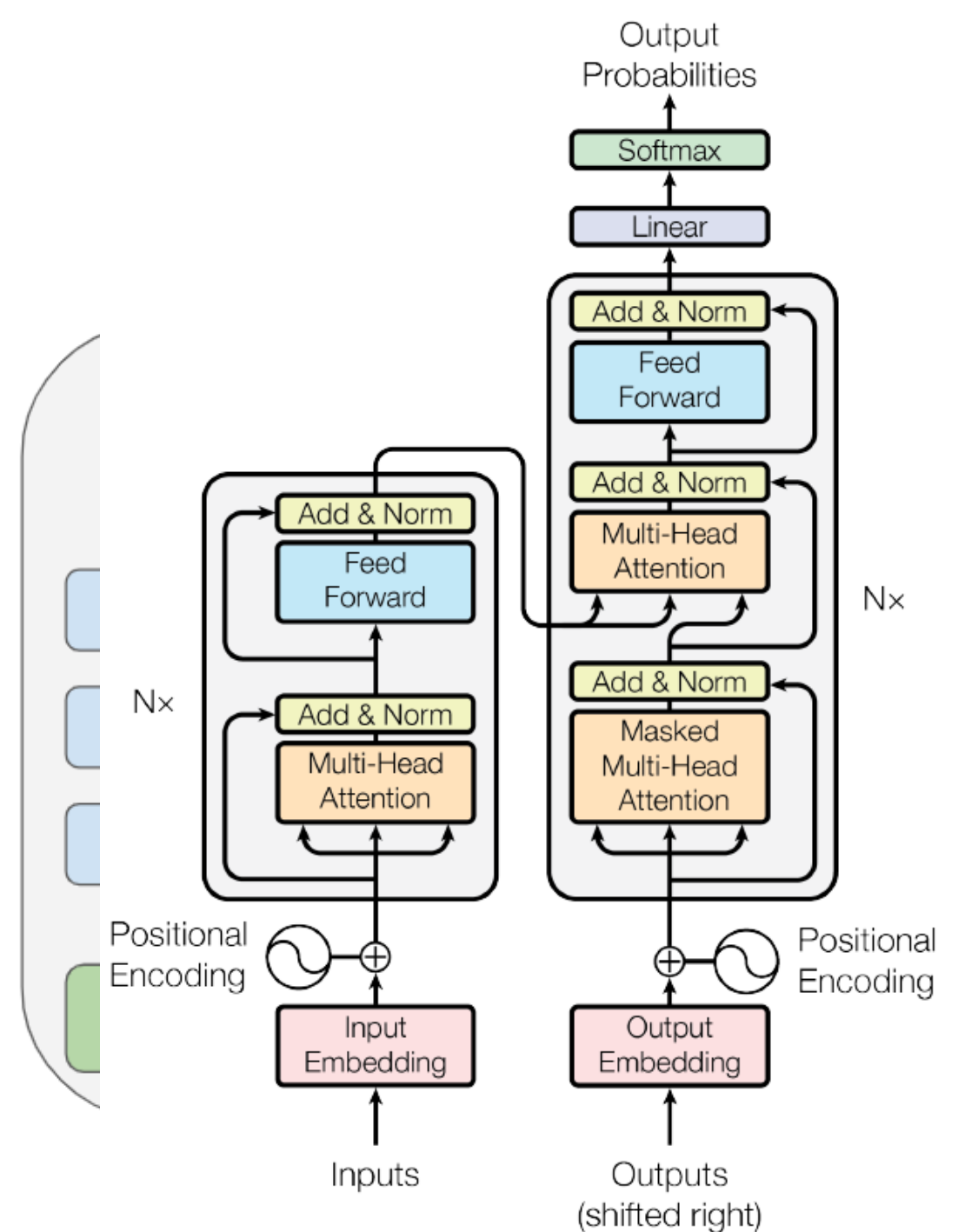
Transformers

Any position can directly attend to any other position, regardless of distance — and every position can be computed in parallel during training.

“Attention Is All You Need” (2017) — the paper that dropped recurrence entirely and built a model on attention alone.

Overview of Architecture

- Transformers are encoder/decoder architectures that were introduced for sequence-to-sequence tasks in NLP.
- The input is given by $X = \{x_1, x_2, \dots, x_T\}$ and the output is $\{y_1, y_2, \dots, y_K\}$.
- In the diagram, the shifted output is what has been decoded produced so far.
- The architecture aims to model $p(y_t | y_{1:t-1}, X)$



Self-Attention: The Core Idea

Every position looks at every other position, and decides how relevant each one is



Weighing Relevance

Each word computes a relevance score against every other word in the sequence — not just its immediate neighbors.



Computed Fresh, Every Time

The weights aren't fixed rules — they're computed dynamically based on the specific input in front of the model.



Direct, Not Chained

Word 1 and word 50 can relate directly in a single step — no need to pass information through 49 intermediate steps.

In short: attention lets the model ask, for every word, “which other words actually matter for understanding this one — right now?”

Self-Attention: A Worked Example

The classic case — figuring out what “it” refers to

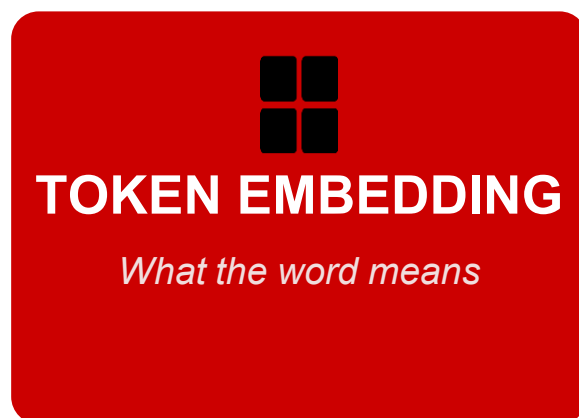


Self-attention lets the model weigh “animal” far more heavily than “street” when interpreting “it” — using the surrounding context, not word order or distance.

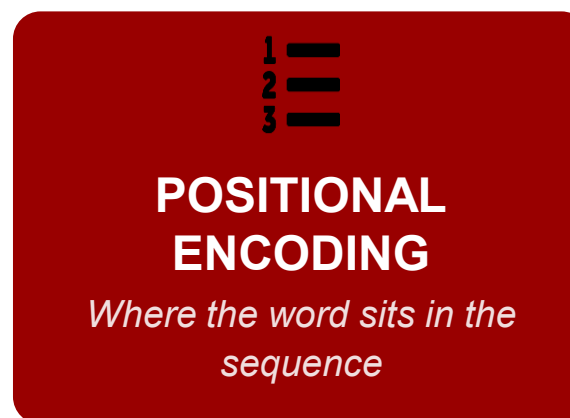
Swap “tired” for “wide” and the weights flip: “it” now points to “street.” Same mechanism, different input, different answer.

Positional Encoding: Giving Attention a Sense of Order

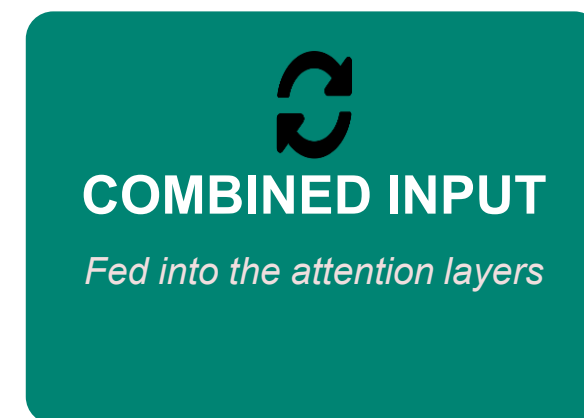
Attention alone has no idea which word came first



+



=



Unlike an RNN, attention treats the input as a set, not a sequence — without this addition, “dog bites man” and “man bites dog” would look identical to it.

The Scale of State-of-the-Art Models

The parameter counts behind this architecture have grown by orders of magnitude

GPT-2 (2019)

1.5B

parameters

GPT-3 (2020)

175B

parameters — ~117× GPT-2

Open MoE models (2026)

1T+

*total parameters (e.g., trillion-parameter
mixture-of-experts models)*

Important nuance: in a mixture-of-experts model, only a fraction of those parameters are “active” for any given token — a trillion-parameter model isn’t doing a trillion-parameter computation on every word.

Most current frontier labs (OpenAI, Anthropic, Google) no longer disclose exact parameter counts — open-weight models are now the best public window into current scale.

Diffusion

A photograph of a modern, multi-story building at dusk. The building features a large glass facade on the right side, which reflects the vibrant orange and pink colors of the sunset sky. The rest of the building is clad in dark, vertical slats. Several windows are illuminated from within, showing a warm yellow light. The sky is filled with scattered clouds, some of which are lit up by the low sun. In the foreground, there is a paved area with some benches and a few small trees. The overall atmosphere is serene and architectural.

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The Gap Diffusion Models Fill

Two earlier approaches to image generation, and their tradeoffs



GANs & Plain VAEs

GANs pit two networks against each other — famously unstable to train. Plain VAEs train stably but tend to produce blurry, averaged-out images.



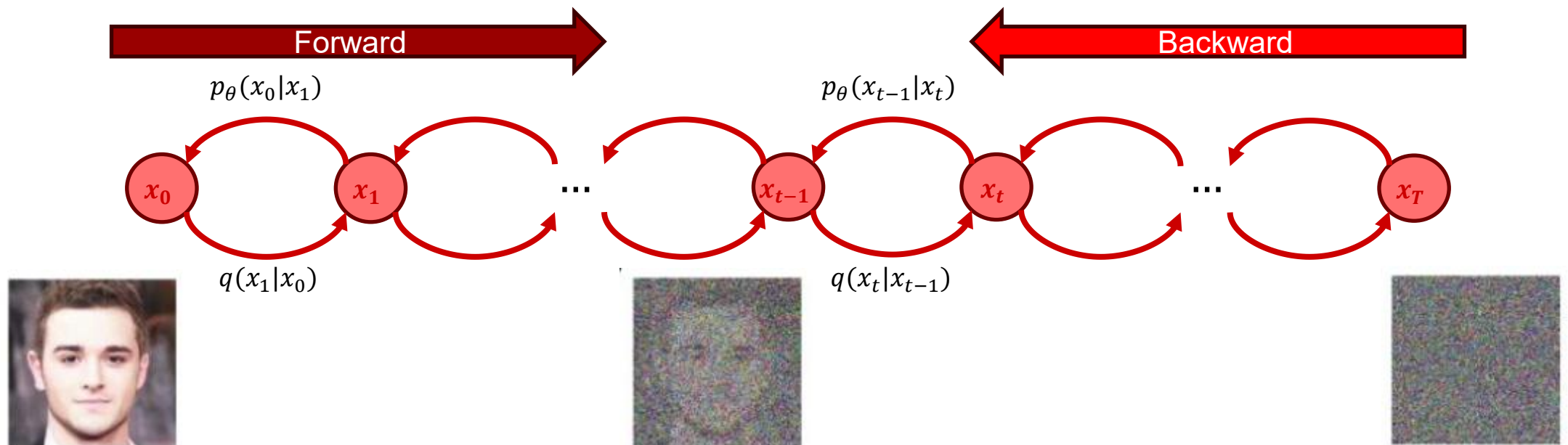
Diffusion Models

A simple, stable training objective — predict the noise — with no adversary to destabilize it. The cost: generation takes many small steps instead of one shot.

Recall from the autoencoder session: a VAE that samples its latent space is already a generative model — diffusion is the far more capable descendant of that same idea.

Motivation

- For Diffusion, we consider a stochastic Markovian process in which we are aiming to obtain samples from the real image distribution.
- A simple **forward process** is defined in which we inject Gaussian noise to the states until a Gaussian distribution is obtained.
- The **backward process** reserves each steps aiming to map from a noisy distribution to the original data manifold.



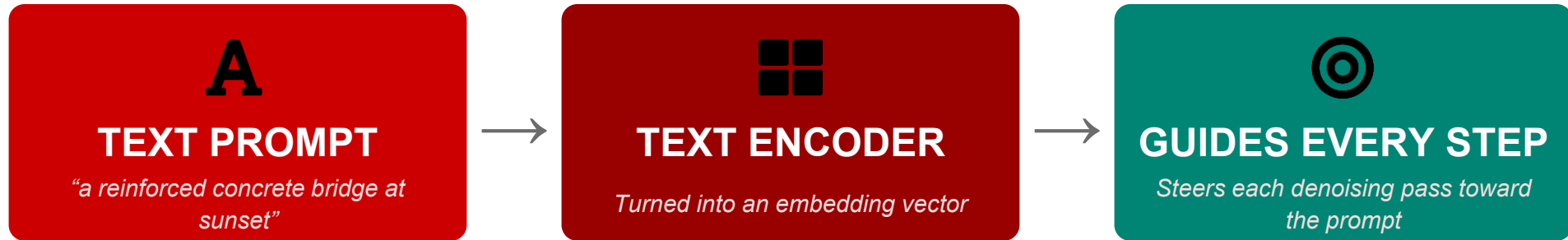
Diffusion Process

- Illustration of Diffusion Process:



Text Conditioning & Guidance

The same prompt steers every single denoising step



Classifier-free guidance controls how strongly the image follows the prompt. Too low and the result drifts off-topic; too high and it becomes oversaturated and artifact-heavy — it's a tunable dial, not a fixed setting.

This is exactly the prompt bar from the LLM session's image-generation slide — same idea, now with the mechanics behind it.

Scale & Applications of State-of-the-Art Models

The parameter counts — and the architecture shift underneath them

SD 1.5 (2022)

0.86B

SDXL (2023)

~2.6B

SD 3.5 / Flux (2024–26)

8–12B

Architecture shift worth noting: the newest models (SD 3.5, Flux) replace the U-Net with a Diffusion Transformer (DiT) — the same attention mechanism from the previous session, now generating images instead of text.

Engineering applications: generative design visualization, and synthetic training images for rare failure modes — both raised back in the classical-ML transition session.